Unit 4

Module 4 : Complex Variables (Differentiation)

5

Complex Variables (Differentiation)

Syllabus

Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; analyticity of elementary functions (exponential, trigonometric, logarithm) and their properties;

Contents

5.1 Introduction

Complex numbers can be defined as ordered pairs (x, y) of real numbers.

In general, a number of the form $z = x + iy$ is called rectangular form of a complex number where x and y are real numbers while $i^2 = -1$.

x is called the real part of z and is denoted by Re(z). y is called the imaginary part of z and is denoted by Im(z).

$x = \text{Re}(z),\ y = \text{Im}(z)$

Thus the set $C = \{z/z = x + iy,\ x, y \in R\}$

represents the set of complex numbers.

Two complex numbers

$z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

are equal whenever

$x_1 = x_2$ and $y_1 = y_2$.

5.2 Fundamental Operation on Complex Number

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers then operations are defined as follows.

i) Addition :

$$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$$
$$= (x_1 + x_2) + i\,(y_1 + y_2)$$

ii) Substraction :

$$z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2)$$
$$= (x_1 - x_2) + i\,(y_1 - y_2)$$

iii) Mutiplication :

$$z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2)$$
$$= (x_1 x_2 - y_1 y_2) + i\,(x_1 y_2 + x_2 y_1)$$

iv) Division :

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{x_1 + iy_1}{x_2 + iy_2} \cdot \frac{x_2 - iy_2}{x_2 - iy_2}$$
$$= \left(\frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2}\right) + i\left(\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}\right)$$

provided $z_2 \neq 0$

Ex. 5.2.1 *Evaluate*

i) (4 +3i) + (–2 – i) *ii) (6 – 8i) – (3i – 9)*

iii) (3 + 2i) (2 – 3i) *iv) (5 – 3i)/(–1 + i)*

Sol. :

i) $(4 + 3i) + (-2 - i) = 4 + 3i - 2 - i$
$$= 2 + 2i$$

ii) $(6 - 8i) - (3i - 9) = 6 - 8i - 3i + 9$
$$= 15 - 11i$$

iii) $(3 + 2i)(2 - 3i) = 3(2 - 3i) + 2i\,(2 - 3i)$
$$= 6 - 9i + 4i - 6i^2$$
$$= 6 - 9i + 4i + 6$$
$$= 12 - 5i$$

iv) $$\frac{(5-3i)}{-1+i} = \frac{5-3i}{-1+i} \times \frac{-1-i}{-1-i}$$
$$= \frac{-5-5i + 3i + 3i^2}{1-i^2}$$
$$= \frac{-8-2i}{2}$$
$$= -4 - i$$

5.3 Conjugate of a Complex Number

Conjugate of a complex number $z = x + iy$ is defined as $x - iy$ and is denoted by $\bar{z}$

Two complex numbers z and $\bar{z}$ are said to be complex conjugate of each other.

Properties :

i) $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$ ii) $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$

iii) $\overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\bar{z}_1}{\bar{z}_2} : z_2 \neq 0$ iv) $\overline{(\bar{z})} = z$

v) $\dfrac{z + \bar{z}}{2} = \text{Re}(z)$ vi) $\dfrac{z - \bar{z}}{2i} = \text{Im}(z)$

5.3.1 Modulus of a Complex Number

If $z = x + iy$ is complex number then modulus or absolute value of z is denoted by $|z|$ and is defined by, $|z| = \sqrt{x^2 + y^2}$

Properties of Modulus

i) $|z_1 \pm z_2| \leq |z_1| + |z_2|$

ii) $|z_1 \pm z_2| \geq \left||z_1| - |z_2|\right|$

iii) $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$

iv) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$

v) $|z| = |\bar{z}|$

vi) $z \cdot \bar{z} = |z|^2$

5.4 Graphical Representation of Complex Number

Complex number x + iy can be considered as an ordered pair of real numbers, We can represent such numbers by points in an xy plane called complex plane or Argand diagram.

For example, $1 + 2i$, $-3 - i$, $-2 + i$, $1 - 3i$ are plot in Fig. 5.4.1.

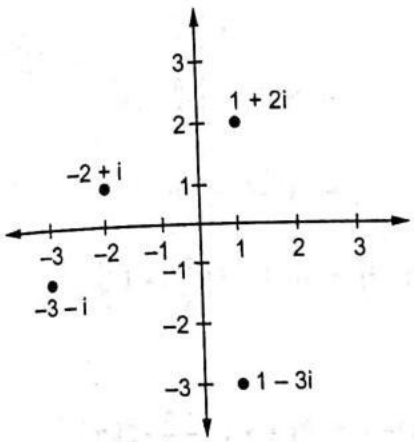

Fig. 5.4.1

The distance between two points $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ in the complex plane is given by,

$$|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

5.4.1 Polar Form of Complex Numbers

The polar form of complex number x + iy is represented as,

$$x + iy = r\cos\theta + i\, r\sin\theta$$

$$= r(\cos\theta + i\sin\theta)$$

where, $r = \sqrt{x^2 + y^2} = |x + iy|$ (modulus or absolute value.)

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$ (Amplitude or principal argument)

Note : Principal argument always lies between

$$-\pi < \theta \leq \pi$$

i) If $x > 0, y > 0$, $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

ii) If $x < 0, y > 0$, $\theta = \pi - \tan^{-1}\left|\frac{y}{x}\right|$

iii) If $x < 0, y < 0$, $\theta = -\pi + \tan^{-1}\left(\frac{y}{x}\right)$

iv) If $x > 0, y < 0$, $\theta = -\tan^{-1}\left|\frac{y}{x}\right|$

Ex. 5.4.1 *Express the following complex numbers in polor form : i)* $1 + \sqrt{3}\, i$, *ii)* $-2 + 2i$, *iii)* $-\sqrt{3} - i$, *iv)* $-2i$

Sol. :

i) $1 + \sqrt{3}\, i$

$$r = \sqrt{1+3} = \sqrt{4} = 2$$

$$\theta = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$1 + \sqrt{3}\, i = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

ii) $-2 + 2i$

$$r = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$$

$$\theta = \pi - \tan^{-1}|1| = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$-2 + 2i = 2\sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$$

iii) $-\sqrt{3} - i$

$$r = \sqrt{3+1} = \sqrt{4} = 2$$

$$\theta = -\pi + \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\pi + \frac{\pi}{6} = \frac{-5\pi}{6}$$

$$-\sqrt{3} - i = 2\left(\cos\frac{5\pi}{6} - i\sin\frac{5\pi}{6}\right)$$

iv) $-2i$

$$r = \sqrt{0+4} = 2$$

$$\theta = -\pi + \tan^{-1}\left(\frac{2}{0}\right) = -\pi + \frac{\pi}{2} = -\frac{\pi}{2}$$

$$-2i = 2\left(\cos\frac{\pi}{2} - i\sin\frac{\pi}{2}\right)$$

5.5 Trigonometric Functions of a Complex Variable z

The trigonometric function of complex variable z are defined as,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\tan z = \frac{\sin z}{\cos z}, \quad \cot z = \frac{\cos z}{\sin z}$$

$$\sec z = \frac{1}{\cos z}, \quad \operatorname{cosec} z = \frac{1}{\sin z}$$

5.6 Hyperbolic Functions

The complex hyperbolic functions are defined by,

$$\cosh z = \frac{e^{z} + e^{-z}}{2}, \quad \sinh z = \frac{e^{z} - e^{-z}}{2}$$

Note : $\cosh(iz) = \dfrac{e^{iz} + e^{-iz}}{2} = \cos z$

$$\sinh(iz) = \frac{e^{iz} - e^{-iz}}{2} = i\,\frac{e^{iz} - e^{-iz}}{2i}$$

$$= i \sin z$$

The general formulae for the real hyperbolic functions continue to hold for the corresponding complex valued functions.

i) $\sinh(z_1 \pm z_2) = \sinh z_1 \cosh z_2 \pm \cosh z_1 \sinh z_2$

ii) $\cosh(z_1 \pm z_2) = \text{coh } z_1 \text{ coh } z_2 \pm \sinh z_1 \sinh z_2$

iii) $\cosh^2(z) - \sinh^2(z) = 1$

iv) $\sinh(2z) = 2\sinh(z)\cosh(z)$

v) $\cosh(2z) = \cosh^2(z) + \sinh^2(z)$

5.7 Logarithm of a Complex Numbers

Let $\quad z = x + iy$

Consider polar co-ordinates

$$x = r\cos\theta,\ y = r\sin\theta$$

So $\quad r = \sqrt{x^2+y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$

$$z = x + iy$$

$$= r\cos\theta + i\,r\sin\theta$$

$$= r(\cos(\theta+2k\pi) + i\sin(\theta+2k\pi))$$

$$z = r\,e^{(\theta+2k\pi)i}$$

$$\ln z = \text{Ln}\, r + i(\theta+2k\pi)$$

Which is general value of logarithm and is denoted by *Ln z*

$\therefore \quad \text{Ln}\, z = \ln r + i(\theta+2k\pi)$

If k = 0, then

$$\ln z = \ln r + i\,\theta$$

which is known as principal value of logarithm of z.

Ex. 5.7.1 *Find general and principal value of logarithm of the following.*

i) i ii) –i iii) 1 + i iv) 1 + $\sqrt{3}$ i

sol. : i) i

$$\ln i = \ln 1 + i\left(\frac{\pi}{2}+2k\pi\right)$$

$$= i\,\frac{i}{2}(\pi+4k\pi)$$

$$\ln(i) = \ln 1 + i\left(\frac{\pi}{2}\right) = i\,\frac{\pi}{2}$$

ii) –i

$$\ln(-i) = \ln 1 + i\left(-\frac{\pi}{2}+2k\pi\right)$$

$$= i\,\frac{1}{2}(4k\pi-\pi)$$

$$\ln(i) = \ln 1 + i\left(-\frac{\pi}{2}\right) = -i\,\frac{\pi}{2}$$

iii) 1 + i

$$\ln(1+i) = \ln\sqrt{2} + i\left(\frac{\pi}{4}+2k\pi\right)$$

$$= \frac{1}{2}\ln 2 + i\left[\frac{1}{4}(8k\pi+\pi)\right]$$

$$\ln(1+i) = \frac{1}{2}\ln 2 + i\,\frac{\pi}{4}$$

iv) $1+\sqrt{3}\,i$

$$\ln(1+\sqrt{3}\,i) = \ln 2 + i\left(\frac{\pi}{3}+2k\pi\right)$$

$$= \ln 2 + i\left[\frac{1}{3}(6k\pi+\pi)\right]$$

$$\ln(1+\sqrt{3}\,i) = \ln 2 + i\,\frac{\pi}{3}$$

5.8 Function of a Complex Variable

GTU : Summer-10, 19

Let H be a set of complex numbers. A function f defined on H is a rule that assigns to each z in s a complex number w. The number w is called the value of f at z and is denoted by f(z), that is w = f(z). The set H is called the domain at definition of f.

Graphical representation :

We know that z = x + iy can be decomposed into real and imaginary part. We find that w = f(z) can also decomposed.

i.e.
$$w = u(x, y) + i\, v(x, y)$$

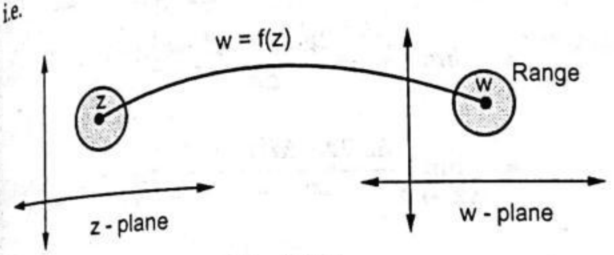

Fig. 5.8.1

Ex. 5.8.1 *Write the function $w = z^2 + i$ in the form u + iv*

Sol. :

$$w = f(z) = z^2 + i$$

$$u + iv = (x+iy)^2 + i \qquad (\because z = x + iy)$$

$$= x^2 + 2ixy - y^2 + i$$

$$= (x^2 - y^2) + i(2xy + 1)$$

So, $u(x, y) = x^2 - y^2$ and $v = 2xy + 1$

Ex. 5.8.2 *Find the range of the function $f(z) = x^2 + i;\ |z| \le 1$.*

Sol. :

$$w = f(z) = u + iv = x^2 + i$$

$$u = x^2 \text{ and } v = 1$$

Given that $|z| \le 1$

$\Rightarrow \quad x^2 + y^2 \le 1$

or $\quad -1 \le x \le 1$

$$0 \le x^2 \le 1$$

$\Rightarrow \quad 0 \le u \le 1, \quad v = 1$

∴ The range at along the line segment from w = i to w = 1 + i.

Ex. 5.8.3 *Find the value of Re(f(z)) and I_m (f(z)) at the indicated point, where.*
$f(z) = \frac{1}{1-z}$ at 7 + 2i

Summer - 10, Marks 3

Sol. :

Here $\quad z = x + iy = 7 + 2i$

$$x = 7 \text{ and } y = 2$$

$$= \frac{1}{1-(x+iy)} = \frac{1}{1-(7+2i)} = \frac{1}{1-7-2i}$$

$$= \frac{1}{-6-2i} = \frac{-(6-2i)}{(6+2i)(6-2i)}$$

$$= -\frac{(6-2i)}{36+4} = -\frac{6}{40} + \frac{1}{20}i$$

$$\text{Rel } f(z) = -\frac{3}{20}, \quad \text{Im}(f(z)) = \frac{1}{20}$$

Ex. 5.8.4 *Write the function $f(z) = z^2 + z + 1$ in the form $f(z) = u(x, y) + iv(x, y)$*

Sol. :

$$u + iv = (x+iy)^2 + (x+iy) + 1$$

$$= x^2 + 2ixy - y^2 + x + iy + 1$$

$$= (x^2 - y^2 + x + 1) + i(2xy + y)$$

$$u = x^2 - y^2 + x + 1,\ v = 2xy + y$$

Ex. 5.8.5 *Find real and imaginary part of $f(z) = z^2 + 4z$. Also, calculate the value of f at z = 1 + i.*

GTU : Summer-19, Maths-4, Marks 4

Sol. :

$$f(z) = z^2 + 4z$$

$$u + iv = (x+iy)^2 + 4(x+iy)$$

$$= x^2 + 2i\,xy - y^2 + 4x + 4iy$$

$$= (x^2 - y^2 + 4x) + i(2xy + 4y)$$

∴ $\quad u = x^2 - y^2 + 4x$ (Real part)

and $\quad v = 2xy + 4y$ (Imaginary part)

$$f(1 + i) = (1+i)^2 + 4(1+i)$$

$$= 1 + 2i - 1 + 4 + 4i$$

$$= 4 + 6i$$

5.9 Polar Form of a Complex Function

The Cartesion of a complex function is

$$f(z) = u_{(x,y)} + iv_{(x,y)} \text{ for } z = x + iy$$

$$\text{Rel } (f(z)) = u_{(x,y)} \text{ and } \text{Im}(f(z)) = v_{(x,y)}$$

Now we put, $x = r\cos\theta$, $y = r\sin\theta$

then $z = x + iy = r\cos\theta + ir\sin\theta$

$$z = re^{i\theta} \ (\because \cos\theta + i\sin\theta = e^{i\theta})$$

with $|z| = r$

and $\theta = \arg(z);$

Thus the polar form at a complex function is

$$f(z) = u(r, \theta) + iv(r, \theta)$$

Ex. 5.9.1 *Write the function $f(z) = z^2 + z + 2$ in the polar form.*

Sol. : $f(z) = z^2 + z + 2$

$$u + iv = (r \cdot e^{i\theta})^2 + r \cdot e^{i\theta} + 2$$

$$= r^2 e^{2i\theta} + r \cdot e^{i\theta} + 2$$

$$= r^2(\cos 2\theta + i\sin 2\theta) + r(\cos\theta + i\sin\theta) + 2$$

$$u + iv = (r^2\cos 2\theta + r\cos\theta + 2) + i(r^2\sin 2\theta + r\sin\theta)$$

$$u = r^2\cos 2\theta + r\cos\theta + 2 \text{ and}$$

$$v = r^2\sin 2\theta + r\sin\theta$$

Ex. 5.9.2 *Find the image in w-plane under the transformation $w = z + 1$ where region at z-plane is $|z| = 1$.*

Sol. : $|z| = 1 \Rightarrow$ radius = 1, Center (0, 0)

$$w = z + 1 \Rightarrow z = |w - 1|$$

$$|w - 1| = 1$$

$$|u + iv - 1| = 1$$

$$|(u - 1) + iv| = 1$$

Center (1, 0) radius 1 in w - plane.

5.10 Complex Derivative

GTU : Summer-10

Let $w = f(z)$ be a complex function of z defined in a domain D. Then f(z) is said to be differentiable $z = z_0$ if the limit

$$f'(z) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Ex. 5.10.1 *By definition P.T. $f(z) = z^2$ is differentiable for all value of z.*

Sol. :

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{(z+\Delta z)^2 - z^2}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{z^2 + 2z\,\Delta z + \Delta z^2 - z^2}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{\Delta z(2z + \Delta z)}{\Delta z}$$

$$\boxed{f'(z) = 2z}$$

So f(z) is differentiable for all value of z.

Ex. 5.10.2 *Find the value of the derivative of $\frac{z-i}{z+i}$ at i.*

Summer-10, Marks 3

Sol. :

$$f(z) = \frac{z - i}{z + i}$$

$$f'(z) = \frac{1(z+i) - 1(z-i)}{(z+i)^2} = \frac{z + i - z + i}{(z+i)^2}$$

$$f'(z)\big|_{z=i} = \frac{2i}{(i+i)^2} = \frac{1}{2i}$$

Ex. 5.10.3 *P.T. $f(z) = \bar{z}$ is not differentiable at any point in the complex plane.*

Sol. :

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{\overline{z + \Delta z} - \bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{\overline{z} + \Delta\overline{z} - \overline{z}}{\Delta z} = \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y}$$

i) Along x-axis, $\Delta y \to 0$

$$= \lim_{\Delta x \to 0} \frac{\Delta x}{\Delta x} = 1$$

ii) Along y-axis, $\Delta x \to 0$

$$= \lim_{\Delta y \to 0} \frac{-i\Delta y}{i\Delta y} = -1$$

So limit does not exist. So $f'(z) = \overline{z}$ is not differentiable at any point of z.

Ex. 5.10.4 By definition P.T. $f(z) = \frac{1}{z}$ is not differentiable at z = 0.

Sol. : $f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$

$$= \lim_{\Delta z \to 0} \frac{\frac{1}{z+\Delta z} - \frac{1}{z}}{\Delta z} = \lim_{\Delta z \to 0} \frac{z - z - \Delta z}{\Delta z\; z(z+\Delta z)}$$

$$= \lim_{\Delta z \to 0} \frac{-\Delta z}{\Delta z\; z(z+\Delta z)} = -\frac{1}{z^2}$$

So $f'(z)$ is not differentiable at z = 0.

5.11 Analytic Functions

GTU : Winter-12,18,19,20,21,22,23,24, Summer-10,18,19,20,21,22,23, Marks 7

A single valued function f(z) is said to be analytic at the point z_0 in the domain D of the z-plane if f(z) is differentiable at $z = z_0$ and also differentiable its neighbourhood at z_0.

An analytic function is also known as holomorphic or regular or monogenic.

Entire function : A function which is analytic any where in the complex plane is said to be entire function.

Theorom : The necessary conditions for f(z) to be analytic.

State : If $f(z) = u_{(x,y)} + i\,v_{(x,y)}$ is differentiable at any point z = x + iy then the real and imaginary parts of u(x, y) and v(x, y) of f(z) are also differentiable at (x, y) and the C - R equation $y_x = v_y$ and $u_y = -v_x$ is satisfied.

Proof : Here given that f(z) is differentiable.

So $f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$ is satisfied.

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$f'(z) = \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \frac{u_{(x+\Delta x,\, y+\Delta y)} + iv_{(x+\Delta x,\, y+\Delta y)} - u_{(x,y)} - iv_{(x,y)}}{\Delta x + i\Delta y}$$

$$(\because f(z) = u_{(x,y)} + iv_{(x,y)}$$

$$f(z+\Delta z) = u_{(x+\Delta x, y+\Delta y)} + iv_{(x+\Delta x, y+\Delta y)}$$

Now, we can take two cases.

1) Along x-axis, $\Delta y \to 0$.

$$f'(z) = \lim_{\Delta x \to 0} \frac{u_{(x+\Delta x, y)} + iv_{(x+\Delta x, y)} - u_{(x,y)} - iv_{(x,y)}}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{u_{(x+\Delta x,y)} - u_{(x,y)}}{\Delta x} + i \lim_{\Delta x \to 0} \frac{v_{(x+\Delta x,y)} - v_{(x,y)}}{\Delta x}$$

$$f'(z) = u_x + iv_x \text{ (By definition)} \quad \dots (A)$$

2) Along y - axis, $\Delta x \to 0$

$$\text{So } f'(z) = \lim_{\Delta y \to 0} \frac{u_{(x,y+\Delta y)} - u_{(x,y)}}{i\Delta y} + i \lim_{\Delta y \to 0} \frac{v_{(x,y+\Delta y)} - v_{(x,y)}}{i\Delta y}$$

$$= -iu_y + v_y \text{ (By definition)} \quad \dots (B)$$

From equations (A) and (B)

$$u_x = v_y \text{ and } u_y = -v_x$$

(Compair real and imaginary part)

Sufficient condition for analytic function

State : If u and v of f(z) are both differentiable at (x, y) and satisfy the C-R equation $u_x = v_y$ and $u_y = -v_x$ then f(z) is differentiable at z = x + iy.

Note 1 : C-R equation are only necessary but not sufficient for a functions to be differentiable at a point.

Note 2 : If f(z) is analytic every where in domain D then f(z) is differentiable every where in D and also C-R equation satisfied in D.

Note 3 : If f(z) is analytic at only $z = z_0$ then f(z) is differentiable only at $z = z_0$ and C-R equation satisfied only at $z = z_0$.

Note 4 : If f(z) is differentiable at $z = z_0$ then it is not analytic at $z = z_0$ but C-R equation satisfied at $z = z_0$.

Note 5 : If C-R equation is not satisfied at z_0 then f(z) is not differentiable at $z = z_0$ and not analytic at $z = z_0$.

Note 6 : If f(z) and g(z) are analytic then $f \pm g$, $f \cdot g$, is analytic and f/g are analytic if $g(z) \neq 0$.

Note 7 : If f(z) is analytic then it is continuous.

Ex. 5.11.1 *Give one example which is not analytic at z = 0 but C-R equation is satisfied at z = 0. or Show that*

$$f(z) = \begin{cases} \sqrt{|xy|}; & z \neq 0 \\ 0; & z = 0 \end{cases}$$

is not analytic at z = 0 but C-R equation is satisifed at z = 0.

Sol. : Here $u = \sqrt{|xy|}$ $v = 0$

$$\frac{\partial y}{\partial x} = \lim_{\Delta x \to 0} \frac{u_{(x+\Delta x, y)} - u_{(x,y)}}{\Delta x}$$

$$\left.\frac{\partial u}{\partial x}\right|_{(0,0)} = \lim_{\Delta x \to 0} \frac{u_{(\Delta x, 0)} - u_{(0,0)}}{\Delta x} = \lim_{\Delta x \to 0} \frac{0-0}{\Delta x} = 0$$

$$\left.\frac{\partial u}{\partial y}\right|_{(0,0)} = \lim_{\Delta y \to 0} \frac{v_{(0,\Delta y)} - v_{(0,0)}}{\Delta y} = 0$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \text{ at } (0, 0)$$

So C - R equation satisfied at z = 0.

Now, we prove that f(z) is not different at z = 0.

$$f'(z) = \lim_{z \to 0} \frac{f(z) - f(0)}{z - 0} = \lim_{z \to 0} \frac{\sqrt{|xy|} - 0}{x + iy}$$

$$= \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \frac{\sqrt{|xy|}}{x + iy}$$

Along x-axis, $y \to 0$

$$\lim_{x \to 0} \frac{0}{x} = 0$$

Along y-axis, $x \to 0$

$$\lim_{y \to 0} \frac{0}{iy} = 0$$

Along y = mx

$$\lim_{x \to 0} \frac{\sqrt{|x\, mx|}}{x + i\, mx} = \lim_{x \to 0} \frac{\cancel{x}\sqrt{|m|}}{\cancel{x}(1 + im)} = \frac{\sqrt{|m|}}{1 + im}$$

So limit does not exist.

So f(z) is not different at z = 0.

Ex. 5.11.2 *Show that*

$$f(z) = \begin{cases} \dfrac{\bar{z}^2}{z} & z \neq 0 \\ 0 & z = 0 \end{cases}$$

C-R equation satisfied at z = 0.

GTU : Winter-18, Summer-20, CVNM, Marks 7

Sol. :
$$f(z) = \frac{\bar{z}^2}{z} = \frac{(x - iy)^2}{x + iy}$$

$$= \frac{(x^2 - 2ixy - y^2)(x - iy)}{(x + iy)(x - iy)}$$

$$= \frac{(x^2 - y^2 - 2ixy)(x - iy)}{x^2 + y^2}$$

$$= \frac{x^3 - x^2 iy - y^2 x + iy^3 - 2ix^2 y - 2xy^2}{x^2 + y^2}$$

$$= \frac{x^3 - 3xy^2}{x^2 + y^2} + i\frac{(y^3 - 3x^2y)}{x^2 + y^2}$$

So $$u = \frac{x^3 - 3xy^2}{x^2 + y^2} \quad v = \frac{y^3 - 3x^2y}{x^2 + y^2}$$

By definition

$$u_x(0, 0) = \lim_{\Delta x \to 0} \frac{u_{(0+\Delta x, 0)} - u_{(0, 0)}}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{\frac{\Delta x^3}{\Delta x^2} - 0}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta x^3}{\Delta x^3} = 1$$

$$u_y(0, 0) = \lim_{\Delta y \to 0} \frac{u_{(0, 0+\Delta y)} - u_{(0, 0)}}{\Delta y} = \lim_{\Delta y \to 0} \frac{0}{\Delta y} = 0$$

$$v_x(0, 0) = \lim_{\Delta x \to 0} \frac{v_{(\Delta x, 0)} - v_{(0, 0)}}{\Delta x} = \lim_{\Delta x \to 0} \frac{0-0}{\Delta x} = 0$$

$$v_y(0, 0) = \lim_{\Delta y \to 0} \frac{v_{(0, \Delta y)} - v_{(0, 0)}}{\Delta y} = \lim_{\Delta y \to 0} \frac{\Delta y}{\Delta y} = 1$$

So, $u_x = v_y$ and $u_y = -v_x$

Satisfied at z = 0.

Ex. 5.11.3 *Examine the nature of function.*

$$f(z) = \begin{cases} \dfrac{x^2 y^5 (x+iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

in the region including the origin.

Sol. : $$f(z) = u + iv = \frac{x^3y^5}{x^4 + y^{10}} + \frac{ix^2y^6}{x^4 + y^{10}}$$

$$u = \frac{x^3y^5}{x^4 + y^{10}}, \quad v = \frac{x^2y^6}{x^4 + y^{10}}$$

$$u_x(0,0) = \lim_{\Delta x \to 0} \frac{u_{(\Delta x, 0)} - u_{(0,0)}}{\Delta x} = \lim_{\Delta x \to 0} \frac{0-0}{\Delta x} = 0$$

$$u_y(0,0) = \lim_{\Delta y \to 0} \frac{u_{(0, \Delta y)} - u_{(0,0)}}{\Delta y} = \lim_{\Delta y \to 0} \frac{0-0}{\Delta y} = 0$$

$$v_x(0,0) = \lim_{\Delta x \to 0} \frac{v_{(\Delta x, 0)} - v_{(0,0)}}{\Delta x} = \lim_{\Delta x \to 0} \frac{0-0}{\Delta x} = 0$$

$$v_y(0, 0) = \lim_{\Delta y \to 0} \frac{v_{(0, \Delta y)} - v_{(0,0)}}{\Delta y} = \lim_{\Delta y \to 0} \frac{0-0}{\Delta y} = 0$$

Hence C - R equation satisfied at origin.

Now we check f(z) is not differential at z = 0.

$$f'(0) = \lim_{z \to 0} \frac{f(z) - f(0)}{z - 0} = \lim_{z \to 0} \frac{\left[\frac{x^2y^5(x+iy)}{x^4 + y^{10}} - 0\right]}{x + iy}$$

$$= \lim_{\substack{x \to 0 \\ y \to 0}} \frac{x^2y^5}{x^4 + y^{10}}$$

i) Along y = mx

$$= \lim_{x \to 0} \frac{x^2m^5x^5}{x^4 + m^{10}x^{10}} = \lim_{x \to 0} \frac{x^3m^5}{1 + m^{10}x^6} = 0$$

ii) Along $y^5 = x^2$

$$= \lim_{x \to 0} \frac{x^2x^2}{x^4 + x^4} = \frac{1}{2}$$

So limit does not exist.

So function is not differential at z = 0.

Ex. 5.11.4 *Verify Cauchy - Rieman equation for* $f(z) = \cos x \cdot \cosh y - i \sin x \cdot \sinh y$.

GTU : Summer-20, CVNM, Marks 3

Sol. :

$$f(z) = \cos x \cdot \cosh y - i \sin x \cdot \sinh y$$

$$u + iv = \cos x \cdot \cosh y - i \sin x \cdot \sinh y$$

$$u = \cos x \cdot \cosh y \qquad v = -\sin x \cdot \sinh y$$

$$y_x = -\sin x \cdot \cosh y \qquad v_x = -\cos x \cdot \sinh y$$

$$u_y = \cos x \cdot \sinh y \qquad v_y = -\sin x \cdot \cosh y$$

$\therefore$ C-R equation : $y_x = v_y$ and $u_y = -v_x$

$\therefore$ C-R equation is satisfied.

Ex. 5.11.5 *Show that* $f(z) = e^{\bar{z}}$ *is nowhere analytic.*

GTU : Summer-22, CVNM, Marks 4

Sol. :

$$f(z) = e^{\bar{z}}$$

$$u + iv = e^{x - iy}$$

$$= e^x(\cos y - i \sin y)$$

$$= e^x \cos y - i\, e^x \cdot \sin y$$

$\therefore$ $u = e^x \cdot \cos y$ $\quad v = -e^x \cdot \sin y$

$u_x = e^x \cdot \cos y$ $\quad v_x = -e^x \cdot \sin y$

$u_y = -e^x \cdot \sin y$ $\quad v_y = -e^x \cdot \cos y$

$\therefore$ C-R equation : $u_x \neq v_y$ and $u_y \neq -v_x$

$\therefore$ C-R equation is not satisfied.

$\therefore$ f(z) is not analytic.

Ex. 5.11.6 *Which is the following function are entire.*

i) $f(z) = 3x + y + i(3y - x)$

ii) $f(z) = xy + iy$ $\quad$ iii) $f(z) = e^y \cdot e^{ix}$

iv) $f(z) = \bar{z} + 1$ $\quad$ v) $f(z) = (z^2 - 1)e^{-x} \cdot e^{-iy}$

vi) $f(z) = z^2 + 2z + 1$ $\quad$ vii) $f(z) = \bar{z}$

GTU : Summer-10,19, Winter-19, 20, CVNM, Marks 3, Summer-23, Marks 4

Sol. :

i) $f(z) = (3x + y) + i(3y - x) = u + iv$

$u = 3x + y$ $\quad v = 3y - x$

$u_x = 3$ $\quad v_x = -1$

$u_y = 1$ $\quad v_y = 3$

So C - R equation is satisfied every where.

u_x, u_y, v_x and v_y continuous every where.

So f(z) is differentiable every where.

So f(z) is analytic every where.

So f(z) is entire function.

ii) $f(z) = xy + iy$

$u = xy$ $\quad v = y$

$u_x = y$ $\quad u_y = x$

$v_x = 0$ $\quad v_y = 1$

So $\quad u_x = v_y$ $\quad u_y = -v_x$

$y = 1$ $\quad x = 0$

So C - R equation satisfied only $x = 0$, $y = 1$ means $z = i$.

u_x, u_y, v_x and v_y continuous every where.

So f(z) is differentiable only $z = i$.

So no where analytic.

So f(z) is not entire function.

iii) $f(z) = e^y \cdot e^{ix}$

$u + iv = e^y(\cos x + i \sin x) = e^y \cos x + ie^y \sin x$

$u = e^y \cos x$ $\quad v = e^y \sin x$

$u_x = -e^y \sin x$ $\quad v_x = e^y \cos x$

$u_y = e^y \cos x$ $\quad v_y = e^y \sin x$

So C - R equation is not satisfied any where.

So f(z) is no where differentiable.

So f(z) is no where analytic.

So f(z) is not entire function.

iv) $f(z) = \bar{z} + 1$

$u + iv = x - iy + 1 = (x + 1) - iy$

$u = x + 1$ $\quad v = -y$

$u_x = 1$ $\quad v_x = 0$

$u_y = 1$ $\quad v_y = -1$

So C - R equation is not satisfied in z - plane.

So f(z) is no where analytic.

So f(z) is not entire function.

v) $f(z) = (z^2 - 1)\, e^{-z}$

Here $\quad f(z) = g(z)\, h(z)$

$g(z) = z^2 - 1$ is analytic any where.

$h(z) = e^{-z}$ is also analytic any where.

So f(z) is analytic any where.

So f(z) is entire function.

vi) $f(z) = z^2 + 2z + 1$

Here $f(z) = z^2 + 2z + 1$ is analytic any where because we know that any polynomial function is analytic.

So f(z) is entire function.

vii) $f(z) = \bar{z}$

$$u + iv = x - iy$$

$$u = x \qquad v = -y$$

$$u_x = 1 \qquad v_x = 0$$

$$u_y = 0 \qquad v_y = -1$$

So C - R equation is not satisfied.

So $f(z) = \bar{z}$ is no where analytic.

So f(z) is not entire function.

Ex. 5.11.7 *Define entire function and show that $f(z) = e^z$ is entire function.*

GTU : Winter-18, Maths-4, Marks 3
Summer-18, CVNM, Marks 3

Sol. : Entire function : A function which is analytic every where in the complex plane is said to be entire function.

$$f(z) = e^z$$

$$\therefore \quad u + iv = e^{x+iy}$$

$$\therefore \quad u + iv = e^x(\cos y + i\sin y)$$

$$\therefore \quad u = e^x \cos y \qquad v = e^x \cdot \sin y$$

$$y_x = e^x \cdot \cos y \qquad v_x = e^x \cdot \sin y$$

$$u_y = -e^x \cdot \sin y \qquad v_y = e^x \cdot \cos y$$

$\therefore$ C-R equation : $u_x = v_y$ and $u_y = -v_x$

$\therefore$ C-R equation satisfied every where u_x, u_y, v_x and v_y continuous every where

$\therefore$ f(z) is analytic every where

$\therefore$ f(z) is entire.

Ex. 5.11.8 *$f(z) = |z|^2$ is analytic ?*

GTU : Winter-22, Maths-4, Marks 3

Sol. : $f(z) = x^2 + y^2$

$$u = x^2 + y^2 \qquad v = 0$$

$$u_x = 2x, \quad u_y = 2y, \quad v_x = 0 \quad v_y = 0$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$$2x = 0 \qquad 2y = 0$$

$$x = 0 \quad \text{and} \quad y = 0$$

So, C - R equation satisfied only x = 0, y = 0, (z = 0).

u_x, u_y v_x and v_y continuous any where

So f(z) is differentiable only z = 0.

So f(z) is no where analytic.

Ex. 5.11.9 *Prove that $f(z) = \frac{1}{z}$ is differentiable every where, except z = 0.*

Sol. :

$$u + iv = \frac{1}{x+iy} = \frac{x-iy}{(x+iy)(x-iy)} = \frac{x-iy}{x^2+y^2}$$

$$= \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$$

$$u = \frac{x}{x^2+y^2} \qquad v = \frac{-y}{x^2+y^2}$$

$$u_x = \frac{1\cdot(x^2+y^2) - x(2x)}{(x^2+y^2)^2} = \frac{y^2+x^2}{(x^2+y^2)^2}$$

$$u_y = 0 - \frac{2xy}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$$

$$v_x = 0 + \frac{2xy}{(x^2+y^2)^2}, \quad v_y = \frac{y^2-x^2}{(x^2+y^2)^2}$$

So, C - R equation satisfied any where.

But, u_x, u_y, v_x and v_y continuous any where except x = 0, y = 0 (means z = 0)

So f(z) is differential any where except z = 0.

So f(z) is analytic any where except z = 0.

But, f(z) is not entire.

Ex. 5.11.10 *Define analytic function. Check analyticity of $f(z) = 2x + iy^2x$.*

GTU : Summer-18, Maths-4, Marks 4

Sol. : Analytic function : A single valued function f(z) is said to be analytic at the point z_0 in the domain D of the z-plane if f(z) is differentiable at $z = z_0$ and also differentiable its neighborhood at z_0.

Here, $\quad f(z) = 2x + i\,xy^2$

$$\therefore \quad u + iv = 2x + ixy^2$$

$$\therefore \quad u = 2x, \quad v = xy^2$$

$$\therefore \quad u_x = 2, \quad v_x = y^2$$

$$u_y = 0, \quad v_y = 2xy$$

$\therefore$ C-R equation

$$u_x = v_y, \quad u_y = -v_x$$

$$2 = 2xy, \quad 0 = -y^2$$

$$\therefore \quad 2 \neq 0 \text{ (Put } y = 0) \qquad \therefore \; y = 0$$

$\therefore$ C-R equation is not satisfied.

$\therefore$ Given function is not analytic.

Ex. 5.11.11 *State necessary and sufficient conditions for the function to be analytic. Also show that f(z) = |z| is not an analytic function.*

GTU : Winter-22, Maths-4, Marks 7

Sol. :

- **Necessary condition :** If f(z) = u + iv is differentiable at any point z = x + iy then the real and imaginary parts of u(x, y) and v(x, y) of f(z) are also differentiable at (x, y) and the C-R equation. $u_x = v_y$ and $u_y = -v_x$ is satisfied.
- **Sufficient condition :** If u and v of f(z) are both differentiable of (x, y) and satisfy the C-R equation $u_x = v_y$ and $u_y = -v_x$, then f(z) is differentiable at z = x + iy.
- Now we prove that f(z) = |z| is not analytic function.

$$f(z) = |z|$$

$$u + iv = \sqrt{x^2 + y^2}$$

$$\therefore \quad u = \sqrt{x^2 + y^2} \quad v = 0$$

$$u_x = \frac{1 \cdot 2x}{2\sqrt{x^2 + y^2}} \quad v_x = 0$$

$$\therefore \quad u_x = \frac{x}{\sqrt{x^2 + y^2}} \quad v_x = 0$$

$$u_y = \frac{y}{\sqrt{x^2 + y^2}} \quad v_y = 0$$

$\therefore$ C-R equation is not satisfied.

$\therefore$ f(z) is not analytic function.

Ex. 5.11.12 *Check whether the following functions are analytic or not at any point*
i) $f(z) = x^2 + ixy$ ii) $f(z) = z^2$

GTU : Summer-19, Maths-4, Marks 4
Summer-22, Marks 4
Winter-22, CVNM, Marks 7

Sol. :

i)
$$f(z) = x^2 + ixy$$

$$u + iv = x^2 + ixy$$

$$u = x^2 \quad \text{and} \quad v = xy$$

$$u_x = 2x \quad v_x = y$$

$$u_y = 0 \quad v_y = x$$

Here, $u_x \neq v_y$ and $u_y \neq -v_x$

$\therefore$ C-R equation is not satisfied.

$\therefore$ f(z) is not analytic.

ii)
$$f(z) = z^2$$

$$u + iv = (x + iy)^2$$

$$u + iv = x^2 + 2ixy - y^2$$

$$u = x^2 - y^2 \quad v = 2xy$$

$$u_x = 2x \quad v_x = 2y$$

$$u_y = -2y \quad v_y = 2x$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$\therefore$ C-R equation is satisfied every where.

$\therefore$ u_x, u_y, v_x and v_y continuous every where.

$\therefore$ f(z) is analytic every where.

Ex. 5.11.13 *Using the C-R equation, show that $f(z) = z^3$ is analytic every where.*

GTU : Winter-20, Maths-4, Marks 3
Summer-21, CVNM, Marks 4
Winter-24, CVPDE, Marks 3

Sol. :

$$f(z) = z^3$$

$$u + iv = (x + iy)^3$$

$$= x^3 + (iy)^3 + 3x^2(iy) + 3x(iy)^2$$

$$= x^3 - iy^3 + i\,3x^2y - 3xy^2$$

$$u + iv = (x^3 - 3xy^2) + i(3x^2y - y^3)$$

$$u = x^3 - 3xy^2 \quad v = 3x^2y - y^3$$

$$\therefore \quad u_x = 3x^2 - 3y^2 \quad v_x = 6xy$$

$$u_y = -6xy \quad v_y = 3x^2 - 3y^2$$

$\therefore$ C-R equation satisfied every where u_x, u_y, v_x and v_y continuous every where.

$\therefore$ f(z) is analytic every where.

Ex. 5.11.14 *Show that $f(z) = \sin z$ is analytic every where.*

GTU : Winter-18, Summer-19, CVNM, Marks 4
Summer-23, CVNM, Marks 3
Winter-24, CVNM, Marks 7

Sol. :

$$f(z) = \sin z$$

$$\therefore \quad u + iv = \sin(x + iy)$$

$$= \sin x \cdot \cos iy + \cos x \cdot \sin iy$$

$$\cos iy = \cosh y \text{ and } \sin iy = i \sinh y$$

$$\therefore \quad u + iv = \sin x \cdot \cosh y + i \cos x \cdot \sinh y$$

$$\therefore \quad u = \sin x \cdot \cosh y \quad v = \cos x \cdot \sinh y$$

$$\therefore \quad u_x = \cos x \cdot \cosh y \quad v_x = -\sin x \cdot \sinh y$$

$$u_y = \sin x \cdot \sinh y \quad v_y = \cos x \cdot \cosh y$$

$\therefore$ C-R equation $u_x = v_y$ and $u_y = -v_x$

$\therefore$ C-R equation is satisfied every where.

u_x, u_y, v_x and v_y is continous every where.

$\therefore$ f(z) is analytic every where.

Ex. 5.11.15 *If $u - v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function at $z = x + iy$ Find $f(z)$ in terms of z.* **GTU : Winter-12, Marks 4**

Sol. :

$$(1 + i) f(z) = (u - v) + i(u + v)$$

$$F(z) = U + iV$$

$$u - v = U = (x - y)(x^2 + 4xy + y^2)$$

$$U_x = 1(x^2 + 4xy + y^2) + (x - y)(2x + 4y)$$

$$U_y = (-1)(x^2 + 4xy + y^2) + (x - y)(4x + 2y)$$

$$U_{x(z,0)} = (z^2) + (z - 0)(2z) = z^2 + 2z^2 = 3z^2$$

$$U_{y(z,0)} = -(z^2) + (z - 0)(4z) = -z^2 + 4z^2 = 3z^2$$

$$F(z) = \int U_{x(z,0)}\,dz - i\int U_{y(z,0)}\,dz$$

$$= \int 3z^2 dz - i\int 3z^2\,dz = z^3 - iz^3 + C$$

$$(1 + i) f(z) = z^3(1 - i) + C$$

$$f(z) = z^3 \frac{(1 - i)}{(1 + i)} + \frac{C}{1 + i} = -iz^3 + C_1$$

Ex. 5.11.16 *If $f(z) = u + iv$ is an analytic function of $z = x + iy$ and $u - v = \dfrac{e^y - \cos x + \sin x}{\cosh y - \cos x}$*

Sol. :

$$(1 + i) f(z) = (u - v) + i(u + v)$$

$$F(z) = U + iV$$

$$(u - v) = U = \frac{e^y - \cos x + \sin x}{\cosh y - \cos x}$$

$$U_x = \frac{+(\sin x + \cos x)(\cosh y - \cos x) - (e^y - \cos x + \sin x)(+\sin x)}{(\cosh y - \cos x)^2}$$

$$U_{x(z,0)} = \frac{(\sin z + \cos z)(1 - \cos z) - (1 - \cos z + \sin z)(\sin z)}{(1 - \cos z)^2}$$

$$= \frac{\cancel{\sin z} - \cancel{\sin z}\cos z + \cos z - \cos^2 z - \cancel{\sin z} + \cancel{\sin z}\cos z - \sin^2 z}{(1 - \cos z)^2}$$

$$= \frac{\cos z - (\cos^2 z + \sin^2 z)}{(1 - \cos z)^2}$$

$$= \frac{\cos z - 1}{(1 - \cos z)^2} = \frac{1}{\cos z - 1}$$

$$U_y = \frac{e^y(\cosh y - \cos x) - (e^y - \cos x + \sin x)(\sinh y)}{(\cosh y - \cos x)^2}$$

$$U_{y(z,0)} = \frac{e^0(1-\cos z)-(e^z-\cos z+\sin z)(0)}{(1-\cos z)^2}$$

$$= \frac{1}{1-\cos z}$$

$$F(z) = \int U_{x(z,0)} dz - i\int U_{y(z,0)} dz$$

$$= \int \frac{1}{\cos z - 1} dz - i\int \frac{1}{1-\cos z} dz$$

$$= -\int \frac{1}{2\sin^2 z/2} dz - i\int \frac{1}{2\sin^2 z/2} dz$$

$$= -\frac{1}{2}\int \text{cosec}^2 z/2\, dz - \frac{i}{2}\int \text{cosec}^2 z/2\, dz$$

$$= -\frac{1}{2}\frac{\cot(z/2)}{1/2} - \frac{i}{2}\frac{\cot z/2}{1/2} + C$$

$$= -(1+i)\cot(z/2) + C$$

$$(1+i)\, f(z) = -(1+i)\cot(z/2) + C$$

$$f(z) = -\cot(z/2) + \frac{C}{(1+i)}$$

$$\boxed{f(z) = \cot(z/2) + C_1}$$

Ex. 5.11.17 *If f(z) = u + iv is an analytic function of z and* $u + v = x^2 - y^2 + 2xy$, *find f(z).*

GTU : Summer-24, Maths-4, Marks 7

Sol. :

$$f(z) = u + iv$$

$$i\, f(z) = iu - v$$

$$\therefore \quad (1+i)\, f(z) = (u-v) + i(u+v)$$

$$\therefore \quad F(z) = U + iV$$

Where, $F(z) = (1+i)\, f(z),\ U = u - v,\ V = u + v$

$$\therefore \quad V = (u+v) = x^2 - y^2 + 2xy$$

By Milne Thomson's method,

$$F(z) = \int V_y(z,0)\, dz + i\int V_x(z,0)\, dz + V(0,0)$$

$$V(0,0) = 0$$

$$V_x = 2x + 2y \Rightarrow V_x(z,0) = 2z$$

$$V_y = -2y + 2x \Rightarrow V_y(z,0) = 2z$$

$$\therefore \quad F(z) = \int 2z\, dz + i\int 2z\, dz$$

$$= z^2 + iz^2 + C$$

$$(1+i)\, f(z) = z^2(1+i) + C$$

$$\therefore \quad f(z) = z^2 + \frac{C}{1+i}$$

Ex. 5.11.18 *If f(z) = u + iv and u - v =* $e^x(\cos y - \sin y)$, *find f(z) in terms of z.*

GTU : Summer-21, Winter-21, Maths-4, Marks 7

Sol. : Now,

$$f(z) = u + iv$$

$$i\, f(z) = i(u+iv) = iu - v$$

$$(1+i)\, f(z) = (u-v) + i(u+v)$$

$$F(z) = U + iV$$

So $U = u - v$ and $V = u + v$

$$U = u - v = e^x(\cos y - \sin y)$$

$$U_x = e^x(\cos y - \sin y)$$

$$U_y = e^x(-\sin y - \cos y)$$

$$U_{x(z,0)} = e^z(\cos 0 - \sin 0) = e^z$$

$$U_{y(z,0)} = -e^z$$

So $F(z) = \int U_{x(z,0)} dz - i\int U_{y(x,0)} dz$

$$= \int e^z dz + i\int e^z dz = e^z + ie^z$$

$$F(z) = (1+i)\, e^z$$

$$(1+i)\, f(z) = (1+i)e^z$$

$$\boxed{f(z) = e^z + C}$$

5.11.1 C - R Equation in Polar Co-ordinate

GTU : Winter-19, Maths-4, Marks 7

Now let $x = r\cos\theta,\ y = r\sin\theta$

$$r = \sqrt{x^2+y^2} \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Now we know that

$$f(z) = u + iv$$

$$z = r\cdot e^{i\theta} \text{ (Using polar co-ordinate)}$$

So $\quad u + iv = f(r - e^{i\theta}) \quad$ ...(A)

Differentiate equation (A) w.r. to r

So $$u_r + i v_r = f'(r\cdot e^{i\theta})\cdot e^{i\theta} \quad \ldots(1)$$

Differentiate equation (A) w.r. to θ

$$u_\theta + i v_\theta = f'(r\cdot e^{i\theta})\cdot r\cdot e^{i\theta}\cdot i \quad \ldots(2)$$

From equations (1) and (2) we get,

$$u_\theta + i v_\theta = \frac{(u_r + i v_r)}{\cancel{e^{i\theta}}}\cdot r\cdot \cancel{e^{i\theta}}\cdot i = i r u_r - r v_r$$

So $$u_\theta = -r v_r, \quad v_\theta = r v_r$$

$$\boxed{\text{So } u_r = \frac{1}{r} v_\theta \text{ and } u_\theta = -r v_r}$$

Ex. 5.11.19 *Which of the following functions are analytic.*

i) $f(z) = z^{1/2}$ ii) $f(z) = z^{3/2}$ iii) $f(z) = \sqrt[3]{r}\, e^{i\frac{\theta}{2}}$

iv) $f(z) = \frac{1}{z^4}$

v) $f(z) = e^{-\theta}\cos(\ln r) + ie^{-\theta}\sin(\ln r)$

GTU : Winter-23, CVNM, Marks 4
Winter-23, CVPPE, Marks 7

Sol. : i) $f(z) = z^{1/2}$

$$u + iv = (r\cdot e^{i\theta})^{1/2} = r^{1/2}\cdot e^{i\theta/2}$$

$$= r^{1/2}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$$

$$u = r^{1/2}\cos\frac{\theta}{2} \quad v = r^{1/2}\sin\frac{\theta}{2}$$

$$u_r = \frac{1}{2\sqrt{r}}\cos\frac{\theta}{2} \quad v_r = \frac{1}{2\sqrt{r}}\sin\frac{\theta}{2}$$

$$u_\theta = \frac{-1}{2} r^{\frac{1}{2}}\sin\frac{\theta}{2} \quad v_\theta = \frac{1}{2} r^{\frac{1}{2}}\cos\frac{\theta}{2}$$

$u_r = \frac{1}{r} v_\theta$	$u_\theta = -r v_r$
$\frac{1}{2\sqrt{r}}\cos\frac{\theta}{2} = \frac{1}{r}\frac{1}{2} r^{1/2}\cos\frac{\theta}{2}$	$\frac{-1}{2} r^{1/2}\sin\frac{\theta}{2} = \frac{-r}{2\sqrt{r}}\sin\frac{\theta}{2}$

C - R equation satisfied any where.

u_r, u_θ, v_r and v_θ continuous every where except $r = 0$ or $z = 0$.

So f(z) is differentiable every where except z = 0.

So f(z) is analytic every where except z = 0.

ii) $f(z) = z^{3/2}$

$$u + iv = (r\cdot e^{i\theta})^{3/2} = r^{\frac{3}{2}} e^{\frac{i3\theta}{2}}$$

$$= r^{3/2}\left(\cos\frac{3\theta}{2} + i\sin\frac{3\theta}{2}\right)$$

$$u = r^{3/2}\cos\frac{3\theta}{2}$$

$$v = r^{3/2}\sin\frac{3\theta}{2}$$

$$u_r = \frac{3}{2} r^{1/2}\cos\frac{3\theta}{2}$$

$$v_r = \frac{3}{2} r^{1/2}\sin\frac{3\theta}{2}$$

$$u_\theta = -\frac{3}{2} r^{3/2}\sin\frac{3\theta}{2},$$

$$v_\theta = \frac{3}{2} r^{3/2}\cos\frac{3\theta}{2}$$

Here $u_r = \frac{1}{r} v_\theta$ and $u_\theta = -r v_r$

So C-R equation satisfied any where. u_r, u_θ, v_r and v_θ continuous any where.

So f(z) is differentiable any where.

So f(z) is analytic any where.

iii) $$f(z) = r^{\frac{1}{3}} e^{\frac{i\theta}{2}}$$

$$u + iv = r^{\frac{1}{3}}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$$

$$u = r^{\frac{1}{3}}\cos\frac{\theta}{2}$$

$$v = r^{\frac{1}{3}}\sin\frac{\theta}{2}$$

$$u_r = \frac{1}{3} r^{\frac{-2}{3}}\cos\frac{\theta}{2}$$

$$v_r = \frac{1}{3} r^{\frac{-2}{3}}\sin\frac{\theta}{2}$$

$$u_\theta = -\frac{1}{2} r^{\frac{1}{3}}\sin\frac{\theta}{2}$$

$$v_\theta = \frac{1}{2} r^{\frac{1}{3}}\cos\frac{\theta}{2}$$

$u_r \neq \frac{1}{r} v_\theta$ and

$$u_\theta \neq -rv_r$$

So C - R equation not satisfied.

So f(z) is no where analytic.

iv) $$f(z) = \frac{1}{z^4}$$

$$u + iv = z^{-4} = (r \cdot e^{i\theta})^{-4} = r^{-4} e^{-4i\theta}$$

$$= r^{-4}(\cos 4\theta - i \sin 4\theta)$$

$$u = r^{-4} \cos 4\theta$$

$$v = -r^{-4} \sin 4\theta$$

$$u_r = -4r^{-3} \cos 4\theta$$

$$v_r = 4r^{-3} \sin 4\theta$$

$$u_\theta = -4r^{-4} \sin 4\theta$$

$$v_\theta = -4r^{-4} \cos 4\theta$$

Here $u_r = \frac{1}{r} v_\theta$ and

$$u_\theta = -rv_r$$

So C - R equation satisfied any where.

But u_r, u_θ, v_r and v_θ is not continuous at $r = 0$ (means $z = 0$)

So f(z) is not differentiable only at z = 0.

So f(z) is analytic any where except z = 0.

v) $$f(z) = e^{-\theta} \cos(ln\, r) + i\, e^{-\theta} \sin(ln\, r)$$

$$u + iv = e^{-\theta} \cos(ln\, r) + i\, e^{-\theta} \sin(ln\, r)$$

$$u = e^{-\theta} \cos(ln\, r)$$

$$v = e^{-\theta} \sin(ln\, r)$$

$$u_r = -e^{-\theta} \sin(ln\, r) \cdot \frac{1}{r}$$

$$u_\theta = -e^{-\theta} \cos(ln\, r)$$

$$v_r = e^{-\theta} \cos(ln\, r) \cdot \frac{1}{r}$$

$$v_\theta = -e^{-\theta} \sin(ln\, r)$$

So $$u_r = \frac{1}{r} v_\theta \text{ and } u_\theta = -r\, v_r$$

Satisfied any where and u_r, u_θ, v_r and v_θ continuous any where .

So f(z) is differentiable any where.

So f(z) is analytic any where.

Ex. 5.11.20 *State Cauchy-Rieman theorem. Write C-R equation in polar-form and verify it for* $f(z) = \frac{z}{\bar{z}}$ *in polar form.* **GTU : Summer-18, CVNM, Marks 7**

Sol. : C-R equation in polar form.

$$u_r = \frac{1}{r} V_\theta \quad \text{and} \quad u_\theta = -r\, V_r$$

$$\therefore \quad f(z) = \frac{z}{\bar{z}}, \; z = r \cdot e^{i\theta}, \; \bar{z} = r \cdot e^{-i\theta}$$

$$u + iv = \frac{r \cdot e^{i\theta}}{r \cdot e^{-i\theta}}$$

$$= e^{2i\theta}$$

$$u + iv = \cos 2\theta + i \sin 2\theta$$

$$u = \cos 2\theta \quad v = \sin 2\theta$$

$$u_r = 0 \quad v_r = 0$$

$$u_\theta = -2 \sin 2\theta \quad v_\theta = 2 \cos 2\theta$$

Here, $u_r \neq \frac{1}{r} v_\theta$ and $u_\theta \neq -r\, v_r$

$\therefore$ C-R equation is not satisfied.

5.12 Harmonic Function

GTU : Winter-18,19,20,21,22,23,24, Summer-18,19,20,21,22,23,24, Maths-4, Marks 3

A real valued function H at two variable x and y is said to be a Harmonic if H is satisfied Laplace equation.

$$H_{xx} + H_{yy} = 0$$

or $$H_{xx}(x, y) + H_{yy}(x, y) = 0$$

Note : If a function $f(z) = u_{(x,y)} + iv_{(x,y)}$ is analytic in a domain D then its component functions u and v are harmonic in D.

Note : A function f(z) = u + iv is analytic in a domain D iff v is a harmonic conjugate of u.

Ex. 5.12.1 *Define harmonic function. Show that $u(x, y) = 2x - x^3 + 3xy^2$ is harmonic in some domain.*

GTU : Summer-18, 19, Maths-4, Marks 3

Sol. : **Harmonic function :** A real valued function ϕ at two variable x and y is said to be a Harmonic if ϕ is satisfied Laplace equations.

$$\phi_{xx} + \phi_{yy} = 0$$

Now, $u(x, y) = 2x - x^3 + 3xy^2$

$$u_x = 2 - 3x^2 + 3y^2$$

$\therefore$ $u_{xx} = -6x$ ...(1)

and $u_y = 6xy$

$u_{yy} = 6x$ ...(2)

From (1) and (2), we write,

$$u_{xx} + u_{yy} = -6x + 6x = 0$$

$\therefore$ u is Harmonic.

Ex. 5.12.2 *Show that $f(x, y) = x^2 - y^2$ is Harmonic function.*

GTU : Winter-18,22, Marks 3
Summer-18, CVNM, Marks 2
Winter-24, CVNM, Marks 3
Summer-24, CVPDE, Marks 2

Sol. : $f(x, y) = x^2 - y^2$

$\therefore$ $f_x = 2x \Rightarrow f_{xx} = 2$,

and $f_y = -2y \Rightarrow f_{yy} = -2$

Now, $f_{xx} + f_{yy} = 2 - 2 = 0$

$\therefore$ f(x, y) is Harmonic function.

Ex. 5.12.3 *Show that $u = y^3 - 3x^2y$ is Harmonic.*

GTU : Summer-21, Winter-18, Maths-4, Marks 3

Sol. : $u = y^3 - 3x^2y$

$u_x = -6xy \quad u_y = 3y^2 - 3x^2$

$u_{xx} = -6y \quad u_{yy} = 6y$

$\therefore$ $u_{xx} + u_{yy} = -6y + 6y = 0$

$\therefore$ u is Harmonic function.

Ex. 5.12.4 *Show that $u = 2x - x^3 + 3xy^2$ is Harmonic in some domain.*

GTU : Summer-22, 24, Marks-4, Marks 3,
Summer-18,19, Winter-19,21, Marks 3,
Summer-20, 21, CVNM, Marks 3,
Winter-22, CVPDE, Marks 3,
Summer-23, VPDE, Marks 3

Sol. : $u = 2x - x^3 + 3xy^2$

$u_x = 2 - 3x^2 + 3y^2$

$\therefore$ $u_{xx} = -6x$

and $u_y = 6xy$

$\therefore$ $u_{yy} = 6x$

Now, $u_{xx} + u_{yy} = -6x + 6x = 0$

$\therefore$ u is Harmonic function.

Ex. 5.12.5 *Show that $u = x \sin x \cdot \cosh y - y \cos x \cdot \sinh y$ is harmonic.*

GTU : Summer-22, CVNM, Marks 4

Sol. : $u = x \cdot \sin x \cdot \cosh y - y \cdot \cos x \cdot \sinh y$

Now, $u_x = \sin x \cdot \cosh y + x \cos x \cdot \cosh y + y \sin x \cdot \sinh y$

$u_{xx} = \cos x \cdot \cosh y + \cos x \cdot \cosh y - x \sin x \cdot \cosh y + y \cos x \cdot \sinh y$ ...(1)

and $u_y = x \sin x \cdot \sinh y - \cos x \cdot \sinh y - y \cos x \cdot \cosh y$

$u_{yy} = x \sin x \cosh y - \cos x \cdot \cosh y - \cos x \cdot \cosh y - y \cos x \cdot \sinh y$ ... (2)

From (1) and (2),

$$u_{xx} + u_{yy} = 0$$

$\therefore$ u is Harmonic.

Examples :

Ex. 5.12.6 *Prove that $u_{(x,y)} = e^x \cos y$ is harmonic. Determine harmonic conjugate and the analytic function $f(z) = u + iv$.*

GTU : Winter-23, CVNM, Marks 4
Winter-23, CVPDE, Marks 7

Sol. :

$$u_x = e^x \cos y$$

$$u_y = -e^x \sin y$$

$$u_{xx} = e^x \cos y$$

$$u_{yy} = -e^x \cos y$$

$$u_{xx} + u_{yy} = 0$$

So u is harmonic.

Now we find $v_{(x,y)}$.

Here, f(z) is analytic function.

So C - R equation satisfied.

$$u_x = v_y \text{ and } u_y = -v_x$$

$$u_x = e^x \cos y = v_y$$

$$u_y = -e^x \sin y = -v_x$$

$$v_x = e^x \sin y \quad \ldots(1)$$

Integrating equation (1) w.r. to x.

$$v_{(x,y)} = e^x \sin y + \phi(y) \quad \ldots(2)$$

Now differentiate equation (2) w.r. to y.

$$v_y = +e^x \cos y + \phi'(y) \quad \ldots(3)$$

We know $v_y = e^x \cos y$ ...(4)

So from equations (3) and (4) (compair 3 and 4)

We get $\phi'(y) = 0$

So $\phi(y) = c$

So $\boxed{V_{(x,y)} = e^x \sin y + C}$

Now $f(z) = u + iv = e^x \cos y + i(e^x \sin y + c)$.

Ex. 5.12.7 *Find a harmonic conjugate of $u(x,y) = 2x - x^3 + 3xy^2$*

GTU : Summer-18,19, Maths-4, Marks 4
Winter-18, Maths-4, Marks 4
Winter-19,21, Summer-22,24, Marks 4
Summer-20,21, CVNM, Marks 4,
Summer-23, CVPDE, Marks 4
Winter-23, CVPDE, Marks 4

Sol. : $u(x, y) = 2x - x^3 + 3xy^2$

$$u_x = 2 - 3x^2 + 3y^2$$

$$u_y = 6xy$$

Here, C-R equation is satisfied.

$$\therefore \quad v_y = 2 - 3x^2 + 3y^2 \quad \ldots(1)$$

$$v_x = -6xy \quad \ldots(2)$$

Now integrating equation (1) w.r. to y.

$$\therefore \quad v(x, y) = 2y - 3x^2y + y^3 + \phi(x) \quad \ldots(3)$$

Now differentiate equation (3) w.r. to x.

$$v_x = -6xy + \phi'(x) \quad \ldots(4)$$

Compair equation (2) and (4),

$$-6xy = -6xy + \phi'(x)$$

$$\therefore \quad \phi'(x) = 0$$

$$\therefore \quad \phi(x) = C$$

$$\therefore \quad v(x, y) = 2y - 3x^2y + y^3 + C$$

Ex. 5.12.8 *Show that the function $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is Harmonic. Find the conjugate harmonic function v.*

GTU : Winter-22, Maths-4, Marks 7
Winter-19, CVNM, Marks 7
Summer-19, CVNM, Marks 4

Sol. : Here, $u(x, y) = 3x^2y + 2x^2 - y^3 - 2y^2$

$$\therefore \quad u_x = 6xy + 4x$$

$$u_{xx} = 6y + 4 \quad \ldots(1)$$

and $u_y = 3x^2 - 3y^2 - 4y$

$$u_{yy} = -6y - 4 \quad \ldots(2)$$

From (1) and (2),

$$u_{xx} + u_{yy} = 6y + 4 - 6y - 4 = 0$$

$\therefore$ u is Harmonic.

Now, C-R equation $u_x = v_y$ and $u_y = -v_x$

$$v_y = 6xy + 4x \quad ...(3)$$

$$\therefore \quad v_x = -3x^2 + 3y^2 + 4y \quad ...(4)$$

Integrating equation (3) w.r. to y.

$$v(x, y) = \frac{6xy^2}{2} + 4xy + \phi(x)$$

$$v(x, y) = 3xy^2 + 4xy + \phi(x) \quad ...(5)$$

Differentiate equation (5) w.r. to x.

$$v_x = 3y^2 + 4y + \phi'(x) \quad ...(6)$$

Compair equation (4) and (6),

$$\therefore \; -3x^2 + 3y^2 + 4y = 3y^2 + 4y + \phi'(x)$$

$$-3x^2 = \phi'(x)$$

$$\therefore \quad \phi(x) = \frac{-3x^3}{3} + C$$

$$\therefore \quad v(x, y) = 3xy^2 + 4xy - x^3 + C$$

Ex. 5.12.9 *Show that $u_{(x,y)}$ is harmonic in some domain D and find harmonic conjugate $v_{(x,y)}$. when*
i) $u_{(x,y)} = y^3 - 3x^2y$ ii) $u_{(x,y)} = 2x(1 - y)$

GTU : Winter-18,19, Summer-19,21, Maths-4, Marks 4

Sol. :

i) $u_{(x,y)} = y^3 - 3x^2y$

$$u_x = -6xy \qquad u_y = 3y^2 - 3x^2$$

Here C - R equation is satisfied

$$v_y = -6xy \quad ...(1)$$

$$v_x = 3x^2 - 3y^2 \quad ...(2)$$

Now integrating equation (1) w.r. to y

$$v_{(x,y)} = \frac{-6xy^2}{2} + \phi(x) = -3xy^2 + \phi(x) \quad ...(3)$$

Differentiate equation (3) w.r. to x

$$v_x = -3y^2 + \phi'(x) \quad ...(4)$$

Compair equations (2) and (4).

$$3x^2 - 3y^2 = -3y^2 + \phi'(x)$$

We get $\quad \phi'(x) = 3x^2$

$$\phi(x) = \frac{3x^3}{3} + C$$

So $\quad v_{(x,y)} = -3xy^2 + x^3 + C$

$$f(z) = u + iv$$

$$= (y^3 - 3xy) + i(-3xy^2 + x^3 + C)$$

ii) $u_{(x,y)} = 2x(1 - y)$

Now,

$$u_x = 2(1-y) \qquad u_y = -2x$$

$$u_{xx} = 0 \qquad u_{yy} = 0$$

So $\quad u_{xx} + u_{yy} = 0$

So u is harmonic.

C - R equation satisfied.

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$$v_y = 2(1 - y) \; ...(1) \qquad v_x = 2x \quad ...(2)$$

Integrating equation (1) w.r.t. y

$$v_{(x,y)} = 2y - y^2 + \phi(x) \quad ...(3)$$

Differentiate equations (3) w.r.to x

$$v_x = 0 + \phi'(x) \quad ...(4)$$

Compair equations (2) and (4)

$$2x = \phi'(x)$$

So $\quad \phi(x) = \frac{\cancel{2}x^2}{\cancel{2}} + C$

So. $\quad \boxed{V_{(x,y)} = 2y - y^2 + x^2 + C}$

Ex. 5.12.10 *Given that $u(x, y) = x^2 - y^2$ and $v(x, y) = -\left(\frac{y}{x^2 + y^2}\right)$. Prove that u and v is Harmonic function but u + iv is not an analytic function.*

GTU : Winter-23, Maths-4, Marks 7

Sol. : First, we prove that u and v is Harmonic.

$\therefore$ u = $x^2 - y^2$, $y_x = 2x$ and $y_{xx} = 2$

$$u_y = -2y \text{ and } u_{yy} = 2$$

$$\therefore \quad u_{xx} + u_{yy} = 2 - 2 = 0$$

$\therefore$ u is Harmonic.

and $$v = \frac{-y}{x^2+y^2}$$

$$v_x = \frac{y}{(x^2+y^2)^2}\cdot 2x = \frac{2xy}{(x^2+y^2)^2}$$

$$v_{xx} = \frac{2y(x^2+y^2)^2 - 2xy\cdot 2(x^2+y^2)\cdot 2x}{(x^2+y^2)^4}$$

$$= \frac{(x^2+y^2)[2y(x^2+y^2)-8x^2y]}{(x^2+y^2)^4}$$

$$v_{xx} = \frac{2yx^2+2y^3-8x^2y}{(x^2+y^2)^3}$$

$$= \frac{2y^3-6x^2y}{(x^2+y^2)^3} \quad ...(1)$$

Now, $$v_y = \frac{(-1)(x^2+y^2)-(-y)(2y)}{(x^2+y^2)^2}$$

$$= \frac{-x^2-y^2+2y^2}{(x^2+y^2)^2}$$

$$v_y = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\therefore v_{yy} = \frac{(2y)(x^2+y^2)^2-(y^2-x^2)\,2(x^2+y^2)(2y)}{(x^2+y^2)^4}$$

$$= \frac{(x^2+y^2)[2y(x^2+y^2)-4y(y^2-x^2)]}{(x^2+y^2)^4}$$

$$\therefore \quad v_{yy} = \frac{2yx^2+2y^3-4y^3+4yx^2}{(x^2+y^2)^3}$$

$$v_{yy} = \frac{6yx^2-2y^3}{(x^2+y^2)^3} \quad ...(2)$$

From (1) and (2),

$$v_{xx}+v_{yy} = \frac{2y^3-6yx^2}{(x^2+y^2)^3}+\frac{6yx^2-2y^3}{(x^2+y^2)^3} = 0$$

$\therefore$ v is Harmonic.

Now, we check f(z) = u + iv is analytic.

$$u_x = 2x \qquad v_x = \frac{2xy}{(x^2+y^2)^2}$$

$$u_y = -2y \qquad v_y = \frac{y^2-x^2}{(x^2+y^2)^2}$$

Here, C-R equation $u_x = v_y$ and $u_y = -v_x$ is not satisfied.

$\therefore$ $f(z) = u + iv$ is not analytic.

Milne Thomson's method :

Let $$f(z) = u_{(x,y)} + iv_{(x,y)}$$

$$f'(z) = u_x + iv_x$$

Using C - R equation $u_x = v_y$ and $u_y = -v_x$

$$f'(z) = u_x - iv_y$$

$$f'(z) = u_{x(z,0)} - iv_{y(z,0)}$$

Taking both side integration,

$$\int f'(z)dz = \int u_{x(z,0)}dz - i\int u_{y(z,0)}dz$$

Note : If given that real part of f(z) and we want to find analytic function f(z) then we use.

$$f(z) = \int u_{x(z,0)}dz - i\int u_{y(z,0)}dz + u_{(0,0)}$$

Note : If given that imaginary part of f(z) and we want to find analytic function f(z) then we use,

$$f(z) = \int v_{y(z,0)}dz + i\int v_{x(z,0)}dz + v_{(0,0)}$$

Ex. 5.12.11 *Find the analytic function of which real part is $e^{-x}((x^2-y^2)\cos y + 2xy\sin y)$*

Sol. : $$u = e^{-x}((x^2-y^2)\cos y + 2xy\sin y)$$

$$u_{(0,0)} = 0$$

$$u_x = -e^{-x}((x^2-y^2)\cos y + 2xy\sin y) + e^{-x}(2x\cos y + 2y\sin y)$$

$$u_y = e^{-x}(-2y\cos y + (x^2-y^2)(-\sin y) + 2x\sin y + 2xy\cos y)$$

$$u_{x(z,0)} = -e^{-z}(z^2) + e^{-z}(2z)$$

$$u_{y(z,0)} = e^{-z}(0) = 0$$

$$f(z) = \int u_{x(z,0)}dz - i\int v_{y(z,0)}dz$$
$$= \int -e^{-z}(z^2) + e^{-z}(2z)dz - i\int 0\,dz$$
$$= \int e^{-z}(2z - z^2)\,dz$$
$$= [(2z - z^2)(-e^{-z}) - (2-2z)(e^{-z}) + (-2)(-e^{-z})] + C$$
$$= [e^{-z}(-\cancel{2}z + z^2 - \cancel{2} + \cancel{2}z + \cancel{2}) + C]$$
$$f(z) = e^{-z}\cdot z^2 + C$$

Ex. 5.12.12 *Construct the analytic function f(z) when*
i) $u = x^3 - 3xy^2 + 3x + 1$
ii) $u = y^3 - 3x^2y$ *iii)* $u = \sin x \cosh y$

Sol. : i) $u = x^3 - 3xy^2 + 3x + 1 \Rightarrow u_{(0,0)} = 1$

$$u_x = 3x^2 - 3y^2 + 3 \Rightarrow u_{x(z,0)} = 3z^2 + 3$$
$$u_y = -6\,xy \Rightarrow u_{y(z,0)} = 0$$
$$f(z) = \int u_{x(z,0)}dz - i\int u_{y(z,0)}dz$$
$$= \int 3z^2 + 3dz - i\int 0\,dz$$
$$= \frac{3z^3}{\cancel{3}} + 3z + C + u_{(0,0)}$$
$$f(z) = z^{\cancel{3}} + 3z + C + 1$$

ii) $u = y^3 - 3x^2y$

$$u_x = -6xy \Rightarrow u_{x(z,0)} = 0$$
$$u_y = 3y^2 - 3x^2 \Rightarrow u_{y(z,0)} = -3z^2$$
$$f(z) = \int u_{x(z,0)}dz - i\int u_{y(z,0)}dz$$
$$= \int 0\,dz - i\int -3z^2 dz$$
$$= +i(z^3) + C + y_{(0,0)}$$
$$= iz^3 + C$$

iii) $u = \sin x \cosh y$

$$u_x = \cos x\ \cosh y \Rightarrow u_{x(z,0)} = \cos z$$
$$u_z = \sin x\ \sinh y \Rightarrow u_{y(z,0)} = 0$$
$$u_{(0,0)} = 0$$
$$f(z) = \int u_{x(z,0)}dz - i\int u_{y(z,0)}dz$$
$$= \int \cos z\,dz - i\int 0\,dz$$
$$= \sin z + C + u_{(0,0)}$$
$$f(z) = \sin z + C$$

Ex. 5.12.13 *Determine the analytic function f(z) = u + iv whose real part is* $u = x^2 - y^2$.

GTU : Summer-18, Maths-4, Marks 4
Summer-23, Maths-4, Marks 7

Sol. : Here, $u = x^2 - y^2$ is given

$$u = x^2 - y^2 \Rightarrow u(0,0) = 0$$
$$u_x = 2x \Rightarrow u_x(z,0) = 2z$$
$$u_y = -2y \Rightarrow u_y(z,0) = 0$$

Using Milne Thomson's Method,

$$f(z) = \int u_x(z,0)\,dz - i\int u_y(z,0)\,dz + u(0,0)$$
$$= \int 2z\,dz - i\int 0\,dz + 0$$
$$= \frac{2z^2}{2} + C$$
$$f(z) = z^2 + C$$

Ex. 5.12.14 *Show that the function* $u(x, y) = x^4 - 6x^2y^2 + y^4$ *is Harmonic. Also find the analytic function f(z) = u(x, y) + i v(x, y).*

GTU : Winter-23, Maths-4, Marks 7

Sol. : Here, $u = x^4 - 6x^2y^2 + y^4$

$\therefore$ $u_x = 4x^3 - 12xy^2$

$u_{xx} = 12x^2 - 12y^2$ ...(1)

and $u_y = -12x^2y + 4y^3$

$u_{yy} = -12x^2 + 12y^2$ ...(2)

From (1) and (2),

$$u_{xx} + u_{yy} = 12x^2 - 12y^2 - 12x^2 + 12y^2$$

$\therefore$ u is Harmonic.

Now, we find analytic function f(z) using Milne Thomson's Method.

$$f(z) = \int u_x(z,0)\,dz - i\int u_y(z,0)\,dz + u(0,0)$$

$\therefore$ $u(0,0) = 0$

$$u_x = 4x^3 - 12xy^2 \Rightarrow u_x(z,0) = 4z^3$$

$$u_y = -12x^2y+4y^3 \Rightarrow u_y(z,0) = 0$$

$$\therefore \quad f(z) = \int 4z^3 dz - i\int 0\, dz + 0$$

$$= \frac{4z^4}{4} + C$$

$$\therefore \quad f(z) = z^4 + C$$

Ex. 5.12.15 *Determine the analytic function whose real part is $e^{-x}(x\cos y - y\sin y)$, using Milne - Thomson's Method.* **GTU : Winter-24, Maths-4, Marks 7**

Sol. : $u(x, y) = e^{-x}(x\cos y - y\sin y)$

Milne-Thomson's Method,

$$f(z) = \int u_x(z,0)\, dz - i\int u_y(z,0)\, dz + u(0,0)$$

$$\therefore \quad u(0,0) = 0$$

$$u_x = -e^{-x}(x\cos y - y\sin y) + e^{-x}\cos y$$

$$\Rightarrow \quad u_x(z,0) = -e^{-z}(z) + e^{-z}1$$

$$u_y = e^{-x}(x(-\sin y) - (\sin y + y\cos y))$$

$$\Rightarrow \quad u_y(z,0) = e^{-z}(z(0) - (0+0)) = 0$$

$$\therefore \quad f(z) = \int -e^{-z}z + e^{-z}dz - i\int 0\, dz + 0$$

$$= \int e^{-z}(1-z)\, dz$$

$$= (1-z)\int e^{-z}\, dz - \int \left((-1)\int e^{-z}dz\right) dz$$

$$= (1-z)(-e^{-z}) + \int (-e^{-z})\, dz$$

$$= -e^{-z} + ze^{-z} + e^{-z} + C$$

$$f(z) = z \cdot e^{-z} + C$$

Ex. 5.12.16 *If $u = x^2 - y^2$, find the corresponding analytic function $f(z) = u + iv$.*

GTU : Winter-20, Maths-4, Marks 4
Summer-18, CVNM, Marks 2
Winter-23, CVNM, Marks 3
Summer-24, CVPDE, Marks 2

Sol. : $\Rightarrow \quad u = x^2 - y^2$

$$\therefore \quad u_x = 2x \Rightarrow u_x(z,0) = 2z$$

$$u_y = -2y \Rightarrow u_y(z,0) = 0$$

$$u(0,0) = 0$$

Now, using Milne - Thomson's Method,

$$f(z) = \int u_x(z,0)\, dz - i\int u_y(z,0)\, dz + u(0,0)$$

$$= \int 2z\, dz - i\int 0\, dz + 0$$

$$f(z) = \frac{2z^2}{2} + C$$

$$\therefore \quad f(z) = z^2 + C$$

Ex. 5.12.17 *Determine the analytic function whose imaginary part is $e^x(x\cos y - y\sin y)$.*

GTU : Winter-22, Maths-4, Marks 7

Sol. : Here, Important part $v = e^x(x\cos y - y\sin y)$ is given,

Here, we use Milne - Thomson's method,

$$f(z) = \int v_y(z,0)\, dz + i\int v_x(z,0)\, dz + v(0,0)$$

$$v = e^x(x\cos y - y\sin y)$$

$$v(0,0) = 0$$

$$v_x = e^x(x\cos y - y\sin y) + e^x(\cos y)$$

$$\Rightarrow \quad v_x(z,0) = e^z(z) + e^z$$

$$v_y = e^x(-x\sin y - (\sin y + y\cos y))$$

$$\Rightarrow \quad v_y(z,0) = 0$$

$$f(z) = 0 + i\int e^z z + e^z dz + 0$$

$$f(z) = i[ze^z - e^z + e^z + C]$$

$$f(z) = i[ze^z + C]$$

Ex. 5.12.18 *Find an analytic function $f(z) = u + iv$ whose real part is $u = x^4 - 6x^2y^2 + y^4$. Also find the corresponding function v.*

GTU : Summer-22, CVNM, Marks 7
Winter-22, CVNM, Marks 3

Sol. : $\Rightarrow \quad u = x^4 - 6x^2y^2 + y^4$

Here, we use Milne-Thomson's Method,

$$f(z) = \int u_x(z,0)\, dz - i\int u_y(z,0)\, dz + u(0,0)$$

$$u(0,0) = 0$$

$$u_x = 4x^3 - 12xy^2 \Rightarrow u_x(z,0) = 4z^3$$

$$u_y = -12x^2y + 4y^3 \Rightarrow u_y(z,0) = 0$$

$$f(z) = \int 4z^3\,dz - i\int 0\,dz + 0$$

$$f(z) = z^4 + C$$

Now, $u + iv = (x+iy)^4$

$= x^4 + 4x^3(iy) + 6x^2(iy)^2 + 4x(iy)^3 + y^4$

$= x^4 + i4x^3y - 6x^2y^2 - i4xy^3 + y^4$

$= (x^4 - 6x^2y^2 + y^4) + i(4x^3y - 4xy^3)$

$\therefore\ v = 4x^3y - 4xy^3$ and $u = x^4 - 6x^2y^2 + y^4$

Ex. 5.12.19 *Show that $u(x, y) = x^2 - y^2 + x$ is Harmonic. Find the corresponding analytic function $f(z) = u + iv$.*

GTU : Winter-24, CVPDE, Marks 4

Sol. : $\Rightarrow u(x, y) = x^2 - y^2 + x$

$\therefore\ u_x = 2x + 1,\ u_{xx} = 2$

$u_y = -2y,\ \ u_{yy} = -2$

$\therefore\ u_{xx} + u_{yy} = 2 - 2 = 0$

$\therefore$ u is Harmonic.

Now, $u_x(z, 0) = 2z + 1$ | $u(0, 0) = 0$

$u_y(z, 0) = 0$

By Milne-Thomson's Method,

$$f(z) = \int u_x(z,0)\,dz - i\int u_y(z,0)\,dz + u(0,0)$$

$$= \int 2z+1\,dz - i\int 0\,dz + 0$$

$$= \frac{2z^2}{2} + z + C$$

$$f(z) = z^2 + z + C$$

Ex. 5.12.20 *Find the constant a so that $u_{(x,y)} = ax^2 - y^2 + xy$ is harmonic. Find its harmonic conjugate and hence $f(z) = u + iv$*

Sol. : $u = ax^2 - y^2 + xy$

$u_x = 2ax + y \quad u_{xx} = 2a$

$U_y = -2y + x \quad u_{yy} = -2$

So $U_{xx} + U_{yy} = 0$

$2a - 2 = 0$

$(a - 1) = 0 \Rightarrow a = 1$

$u = x^2 - y^2 + xy$

$u_x = 2x - y \quad u_y = -2y + x$

$u_{x(z,0)} = 2z \quad u_{y(z,0)} = z$

$$v = \int 2z\,dz - i\int z\,dz = z^2 - iz^2 + C$$

$$= \left(1 - \frac{i}{2}\right)z^2 + C = \left(1 - \frac{i}{2}\right)(x+iy)^2 + C$$

$$= \left(1 - \frac{i}{2}\right)(x^2 + 2ixy - y^2)^2 + C$$

$$= x^2 + 2ixy - y^2 - \frac{i}{2}x^2 - xy + \frac{i}{2}y^2$$

$$= (x^2 - y^2 - xy) + i\left(2xy - \frac{x^2}{2} + \frac{y^2}{2}\right)$$

Ex. 5.12.21 *If $W = \phi + i\psi$ represent the complex potential for an electric field and $\psi = x^2 - y^2 + \frac{x}{x^2y^2}$ determine the function ϕ.*

Sol. :

$$\psi = x^2 - y^2 + \frac{x}{x^2 - y^2}$$

$$\psi_x = 2x + \frac{(x^2+y^2) - x(2x)}{(x^2+y^2)^2}$$

$$\psi_{x(z,0)} = 2z + \frac{(-z^2)}{x^4} = 2z - \frac{1}{z^2}$$

$$\psi_y = -2y + \frac{(0) - x(2y)}{(x^2+y^2)^2}$$

$$\psi_{y(z,0)} = 0$$

$$f(z) = \int \psi_{y(z,0)}\,dz + i\int \psi_{x(z,0)}\,dz$$

$$= 0 + i\int 2z - \frac{1}{z^2}\,dz = z^2 + \frac{1}{2} + C$$

$$= (x+iy)^2 + \frac{(x-iy)}{x^2+y^2}$$

$$= \frac{(x^2 + 2ixy - y^2)(x^2+y^2) + x - iy}{x^2 - y^2}$$

$$= \frac{x^4 + x^2\cancel{y^2} + 2ix^3y + 2ixy^3 - \cancel{y^2}x^2 - y^4 + x - iy}{x^2+y^2}$$

$$= \frac{(x^4 - y^4) + x}{x^2+y^2} + i\left(\frac{2x^3y + 2xy^3}{x^2+y^2}\right)$$

$$= x^2 - y^2 + \frac{x}{x^2+y^2} + i\left(2xy - \frac{y}{x^2+y^2}\right)$$

$$\phi = 2xy - \frac{y}{x^2+y^2} + C$$

Ex. 5.12.22 *Find the analytic function f (z) whose real part is $u = ax^3 + bxy$ and also find a and b where u is harmonic.* **GTU : Summer-23, Maths-4, Marks 7 Winter-23, CVPDE, Marks 3**

Sol. :

$$u = ax^3 + bxy \Rightarrow \quad u_x = 3ax^2 + by$$

$$u_y = bx \qquad u_{xx} = 6\,ax$$

$$u_{yy} = 0$$

$$u_{xx} + u_{yy} = 6ax + 0 = 0$$

$$\therefore \quad 6ax = 0$$

$$a = 0 \text{ and } b \in \not{C}$$

So $\quad u = bxy, \quad u_x = by, \quad u_y = bx$

$$u_{x(z,0)} = u_{y(z,0)} = bz$$

$$f(z) = \int u_{x(z,0)}\,dz - i\int u_{y(z,0)}\,dz$$

$$= \int 0\,dz - i\int bz\,dz$$

$$\boxed{f(z) = -i\frac{bz^2}{2} + C}$$

Ex. 5.12.23 *Show that on analytic function with constant modulus is constant.*

GTU : Summer-21, Maths-4, Marks 7

Sol. : Let

$$f(z) = u + iv$$

$$|f(z)| = C$$

$$u^2 + v^2 = C \qquad \ldots(1)$$

Differentiate equation (1) w.r.to x

$$2u\frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x} = 0$$

$$2u\frac{\partial u}{\partial y} + 2v\frac{\partial v}{\partial y} = 0$$

Using C-R equation

$$2u\,u_x - 2v\,u_y = 0 \qquad \ldots(2)$$

$$(\because u_x = u_y \text{ and } u_y = -v_x)$$

Multiply equation (1) w.r. to v and

Multiply equation (2) w.r. to u

$$2v\,u_x + 2u\,u_y = 0 \qquad \ldots(3)$$

$$2uv\,\cancel{u_x} - 2v^2u_y = 0 \qquad \ldots(4)$$

$$2\cancel{uv}\,u_x + 2u^2u_y = 0 \qquad \ldots(5)$$

$$- \quad -$$

$$-(2u^2 + 2v^2)\,u_y = 0$$

$$(u^2 + v^2)\frac{\partial u}{\partial y} = 0$$

$$\frac{\partial u}{\partial y} = 0 \Rightarrow \boxed{u = C}$$

Similarly $\quad \frac{\partial v}{\partial x} = 0 \Rightarrow \boxed{v = C}$

So $\quad f(z) = u + iv$ is also constant.

Ex. 5.12.24 *Let a function f(z) be analytic in a domain D. Prove that f(z) must be constant in D in each at the following cases :*

a) If f(z) is real valued for all z in D.

b) If $\overline{f(z)}$ is analytic in D.

Sol. :

a) Here given that f(z) is real valued function for all z in D.

So, $\quad f(z) = u + iv$ but $v = 0$

So, $\quad v_x = 0 \quad v_y = 0$

$f(z) = u + iv$ is analytic

C-R equation is satisfied.

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$$u_x = 0 \text{ and } u_y = 0$$

So,

$$u_x = u_y = 0 \Rightarrow u = C$$

$$v_x = v_y = 0 \Rightarrow v = C$$

b) Here $f(z) = u + iv$ and $\overline{f(z)} = u - iv$ is analytic.

Now we take first

$$f(z) = u + iv \text{ is analytic.}$$

So C-R equation satisfied.

$$u_x = v_y \quad ...(1)$$

and

$$u_y = -v_x \quad ...(2)$$

$$\overline{f(z)} = u - iv \text{ is analytic.}$$

So C-R equation satisfied

$$u_x = -v_y \quad ...(3)$$

$$u_y = v_x \quad ...(4)$$

From equations (1) and (3) we get,

$$u_x + u_x = 0 \Rightarrow 2u_x = 0$$

$$\Rightarrow u_x = 0$$

$$\Rightarrow \boxed{u = C}$$

From equations (2) and (4) we get

$$u_y - u_y = -v_x - v_x$$

$$-2v_x = 0 \Rightarrow v_x = 0 \Rightarrow \boxed{v = C}$$

So, $f(z) = u + iv$ is also constant.

Ex. 5.12.25 *Show that an analytic function in a domain with its derivative zero for every point of the domain is constant.*

Sol. : We have to show that if $f'(z)$ is zero for every point of a domain D, then $f(z)$ is constant in D.

We have for such function,

$$f(z) = u + iv \qquad f'(z) = u_x + iv_x$$

So that $f'(z) = 0$

$\Rightarrow u_x = 0$ and $v_x = 0$

$\Rightarrow v_y = 0$ and $u_y = 0$

$\Rightarrow u_x = 0, u_y = 0 \Rightarrow u$ is constant.

ans $v_x = 0, v_y = 0 \Rightarrow v$ is constant.

Ex. 5.12.26 *If f(z) is regular function (Analytic function) of z then*

Prove that :

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|Rel\,(f(z))|^2 = 2\,|f'(z)|^2$$

GTU : Summer-22, Maths-4, Marks 7

Sol. : We have

$$f(z) = u + iv$$

$$Rel\,(f(z)) = u$$

$$|Rel\, f(z)|^2 = |u^2| = u^2$$

$$\text{L.H.S} = \left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)(u^2)$$

$$= \frac{\partial^2}{\partial x^2}(u^2)+\frac{\partial^2}{\partial y^2}(u^2)$$

$$= \frac{\partial}{\partial x}(2u\,u_x)+\frac{\partial}{\partial y}(2u\,u_y)$$

$$= 2u_x^2 + 2u\,u_{xx} + 2y_y^2 + 2u\,u_{yy}$$

$$= 2u(u_{xx}+u_{yy}) + 2(u_x^2+u_y^2)$$

But we know that $f(z)$ is analytic then $f(z)$ is also harmonic so.

$$u_{xx}+u_{yy} = 0$$

$$= 2u(0)+2(u_x^2+u_y^2) = 2(u_x^2+u_2^2)$$

$$= 2|f'(z)|^2$$

$$(\because \; f(z) = u + iv$$

$$f'(z) = u_x + iv_x$$

$$|f'(z)| = \sqrt{u_x^2+v_x^2} = \sqrt{u_x^2+u_y^2}\;)$$

Ex. 5.12.27 *If f(z) = u + iv is regular function (analytic function) then prove that*

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|I_m(z)|^2 = 2|f'(z)|^2$$

GTU : Winter-23, CVNM, Marks 4
Winter-23, CVPDE, Marks 7

Sol. : $\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)(v^2)$

$$= \frac{\partial^2}{\partial x^2}(v^2)+\frac{\partial^2}{\partial y^2}(v^2)$$

$$= \frac{\partial}{\partial x}(2v\,v_x)+\frac{\partial}{\partial y}(2vv_y)$$

$$= 2v_x^2+2v\,v_{xx}+2v_y^2+2v\,v_{yy}$$

$$= 2v(v_{xx}+v_{yy})+2v_x^2+2v_y^2$$

$$= 2v(0)+2(v_x^2+v_y^2) = 2(v_x^2+v_y^2)$$

$$= 2\,|f'(z)|^2$$

Ex. 5.12.28 *If $f(z) = u + iv$ is analytic function then prove that*

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4\,|f'(z)|^2$$

GTU : Summer-19, Maths-4, Marks 7

Sol. :

$$|f(z)|^2 = f(z)\cdot f(\overline{z})$$

$$= \left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2$$

$$= \left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)(u^2+v^2)$$

$$= \frac{\partial^2}{\partial x^2}(u^2+v^2)+\frac{\partial^2}{\partial y^2}(u^2+v^2)$$

$$= \frac{\partial}{\partial x}(2u\,u_x+2v\,v_x)+\frac{\partial}{\partial y}(2u\,u_y+2v\,v_y)$$

$$= 2u_x^2+2u\,u_{xx}+2v_x^2+2v\,v_{xx} + 2u_y^2+2u\,u_{yy}+2v_y^2+2v\,v_{yy}$$

$$= 2(u_x^2+v_x^2)+2(u_y^2+v_y^2) + 2u\,(u_{xx}+u_{yy})+2v\,(v_{xx}+v_{yy})$$

$$= 2(u_x^2+v_x^2)+2(u_y^2+v_y^2)$$

$$(\because u_{xx}+u_{yy}=0,\quad v_{xx}+v_{yy}=0)$$

$$= 2\,|f'(z)|^2+2\,|f'(z)|^2 = 4\,|f'(z)|^2$$

Ex. 5.12.29 *Prove that* $\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right) = 4\frac{\partial^2}{\partial z\,\partial \overline{z}}$

Sol. : Let $x + iy = z,\quad x - iy = \overline{z}$

$$x = \frac{1}{2}(z+\overline{z}),\quad y = -\frac{i}{2}(z-\overline{z})$$

$$\frac{\partial x}{\partial z} = \frac{1}{2}\ \frac{\partial x}{\partial \overline{z}} = \frac{1}{2}\ \frac{\partial y}{\partial z} = \frac{-i}{2},\ \frac{\partial y}{\partial \overline{z}} = \frac{i}{2}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x}\cdot\frac{\partial x}{\partial z}+\frac{\partial}{\partial y}\cdot\frac{\partial y}{\partial z} = \frac{\partial}{\partial x}\left(\frac{1}{2}\right)+\frac{\partial}{\partial y}\cdot\left(-\frac{i}{2}\right)$$

$$\frac{\partial}{\partial \overline{z}} = \frac{\partial}{\partial x}\cdot\frac{\partial x}{\partial \overline{z}}+\frac{\partial}{\partial y}\cdot\frac{\partial y}{\partial \overline{z}} = \frac{\partial}{\partial x}\left(\frac{1}{2}\right)+\frac{\partial}{\partial y}\left(\frac{i}{2}\right)$$

Now,

$$\frac{\partial^2}{\partial z\cdot\partial \overline{z}} = \frac{\partial}{\partial z}\cdot\frac{\partial}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x}-i\frac{\partial}{\partial y}\right)\frac{1}{2}\left(\frac{\partial}{\partial x}+i\frac{\partial}{\partial z}\right)$$

$$= \frac{1}{4}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)$$

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right) = 4\frac{\partial^2}{\partial z\,\partial \overline{z}}$$

Ex. 5.12.30 *If $w = f(z)$ is a regular function at z prove that*

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\log|f'(z)| = 0$$

Sol. :

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\{\log|f'(z)|\}$$

$$= 4\frac{\partial^2}{\partial z\,\partial \overline{z}}\{\log|f'(z)|$$

$$= 4\frac{\partial^2}{\partial z\,\partial \overline{z}}\left\{\frac{1}{2}\log|f'(z)|^2\right\}$$

$$= 2\frac{\partial^2}{\partial z\,\partial \overline{z}}\{(f'(z)\,\overline{f'(z)})\}$$

$$= 2\frac{\partial^2}{\partial z\,\partial \overline{z}}\{\log f'(z)+\log\overline{f'(z)}\}$$

$$= 2\frac{\partial}{\partial \overline{z}}\left\{\frac{f''(z)}{f'(z)}\right\} = 0$$

□□□